

LONG ISLAND MACARTHUR, NY, USA

WMO: 725050

Lat: 40.794N

Lon: 73.102W

Elev: 26

StdP: 101.02

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04781

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-9.1	-20.6	0.6	-9.4	-18.3	0.8	-7.4	12.8	-1.3	11.8	-1.8	4.7	290	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	31.5	23.1	29.9	22.3	28.5	21.8	25.0	28.6	24.2	27.4	23.5	26.5	4.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.9	26.8	23.1	17.9	25.9	22.5	17.3	25.4	76.0	28.8	72.9	27.5	70.1	26.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.3	8.5	-14.5	34.9	2.3	2.1	-16.1	36.4	-17.5	37.6	-18.8	38.8	-20.4	40.3		
			-15.4	26.3	2.2	1.1	-17.0	27.1	-18.3	27.8	-19.5	28.4	-21.1	29.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-0.1	0.7	4.3	9.9	15.2	20.5	23.9	23.2	19.5	13.3	7.6	3.0
	DBStd	9.29	5.39	4.74	4.48	3.85	3.62	3.41	2.66	2.64	3.27	4.03	4.36	4.67
	HDD10.0	1142	316	262	183	46	2	0	0	0	0	13	97	223
	HDD18.3	2872	573	494	435	253	112	18	1	1	26	163	322	476
	CDD10.0	1799	1	1	7	44	163	315	430	408	284	117	24	5
	CDD18.3	487	0	0	0	1	14	82	172	150	59	7	0	0
	CDH23.3	3185	0	0	1	15	103	493	1284	1003	262	25	0	0
	CDH26.7	813	0	0	0	4	22	126	377	243	37	4	0	0
Wind	WSAvg	4.1	4.6	4.6	4.8	4.6	4.0	3.7	3.6	3.3	3.6	3.9	4.1	4.3
Precipitation	PrecAvg	1178	94	83	112	103	91	97	88	107	100	102	84	117
	PrecMax	1507	214	163	249	183	197	209	170	261	207	375	149	215
	PrecMin	955	49	32	24	40	24	19	37	19	35	11	44	48
	PrecStd	149	35	36	50	44	39	55	36	61	42	74	28	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	14.6	20.1	25.2	29.2	32.5	34.5	32.8	29.9	26.3	19.6	16.8
		MCWB	12.1	9.6	12.6	15.5	19.9	23.1	24.3	24.1	22.7	19.0	14.7	13.2
	2%	DB	12.1	11.5	15.5	21.0	26.3	29.9	32.1	30.9	27.5	23.0	17.6	14.0
		MCWB	10.8	8.5	9.8	13.5	18.0	21.9	23.4	23.1	20.9	19.0	14.6	12.3
	5%	DB	10.0	9.7	12.7	18.0	23.5	27.8	30.2	29.1	26.1	21.4	16.3	12.3
		MCWB	8.2	7.2	8.9	11.4	17.0	20.8	22.7	22.3	20.4	17.8	14.0	10.8
	10%	DB	7.8	7.8	10.9	16.1	21.4	26.2	28.7	27.8	24.7	19.8	14.7	10.5
		MCWB	5.9	5.4	7.7	10.6	15.8	19.9	22.2	21.7	20.1	16.7	12.3	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	11.7	14.4	17.5	21.6	24.5	26.4	26.1	24.4	22.0	17.4	14.7
		MCDB	13.9	12.6	19.0	24.0	27.3	30.0	31.9	30.6	27.2	23.2	18.3	15.8
	2%	WB	11.0	9.6	11.5	15.1	19.9	23.2	25.1	25.1	23.3	20.3	15.9	12.6
		MCDB	11.7	10.9	13.5	18.6	24.0	27.6	29.2	28.1	25.3	22.3	16.9	13.5
	5%	WB	8.6	7.7	9.8	13.5	18.6	22.3	24.3	24.3	22.5	18.9	14.4	11.0
		MCDB	9.6	8.9	11.9	16.1	22.2	25.9	27.8	27.1	24.4	20.6	15.8	12.2
	10%	WB	6.3	5.6	8.0	12.0	17.1	21.3	23.6	23.5	21.5	17.4	12.8	8.8
		MCDB	7.6	7.6	10.1	14.4	20.2	24.5	26.9	26.1	23.6	19.2	14.4	10.1
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	8.4	9.2	9.1	8.6	8.2	8.1	8.5	9.0	8.4	7.6
		MCDBR	9.5	10.0	11.0	12.3	12.2	10.6	9.8	9.2	9.1	10.0	9.1	9.0
		MCWBR	8.6	8.0	7.3	6.6	6.2	4.9	4.0	4.1	4.8	6.4	7.4	8.5
	5% WB	MCDBR	9.1	9.6	10.0	9.7	10.4	8.8	8.0	7.2	7.0	8.2	7.9	8.5
		MCWBR	9.1	9.0	7.9	6.6	5.9	4.9	3.9	3.8	4.6	6.5	7.7	8.9
Clear-Sky Solar Irradiance	taub	0.307	0.314	0.336	0.381	0.431	0.476	0.488	0.467	0.396	0.345	0.317	0.307	
	taud	2.522	2.502	2.483	2.378	2.272	2.185	2.162	2.215	2.385	2.540	2.567	2.567	
	Ebn at Noon	858	905	918	892	850	809	794	801	841	857	840	831	
	Edh at Noon	73	86	98	117	133	145	148	136	108	83	70	66	
All-Sky Solar Radiation	RadAvg	1.84	2.73	3.78	4.86	5.43	5.95	6.02	5.39	4.39	3.02	2.09	1.57	
	RadStd	0.15	0.27	0.32	0.49	0.54	0.57	0.31	0.34	0.35	0.28	0.17	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	+48
Regional (69 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+63

Nomenclature: See separate page